Detectable Output Distribution Shift Under Lie-Inducing Instructions in GPT-4.1

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Abstract

Instruction-conditioned deception in large language models is a practical safety problem, but many prior detectors rely on internal activations that are unavailable in black-box deployments. We ask a simpler question: can output text alone reveal when a model is pushed to lie? We run paired generation with GPT-4.1 on 120 TRUTHFULQA questions, producing truthful and lie-roleplay answers under matched decoding settings. We then test separability with both a text classifier and two-sample distribution tests.

Our output-only classifier (TF-IDF+LR) reaches an AUROC of 0.883 on a held-out question split, with permutation-test $p = 9.99 \times 10^{-4}$ and bootstrap 95% CI $[0.798, 0.952]$. Independent distribution testing also rejects equal-distribution null hypotheses: MMD is 0.02246 with permutation $p = 0.001996$. The signal is robust to seed shift, where AUROC increases to 0.910 on unseen seed-7 generations. Feature-level analysis finds one strong univariate marker (punctuation ratio, FDR-corrected $p = 1.97 \times 10^{-13}$, Cohen's $d = 0.83$), while most predictive signal remains multivariate.

These results establish a reproducible black-box baseline for deception-risk monitoring from text alone. The findings suggest that lie-inducing instructions create stable output-distribution shifts that can be detected without privileged model access.

1 Introduction

Large language models can be pushed into deceptive behavior by roleplay or jailbreak-style instructions. This matters operationally because production systems usually observe only prompts and outputs, not hidden states.

why is this hard in practice? Existing deception detection results are strongest when they probe internal activations [Goldowsky-Dill et al., 2025, Long et al., 2025]. These methods are useful for mechanistic understanding, but they are difficult to deploy in API-only settings. Our main question is whether we can detect deception pressure from text alone.

what gap do we address? Prior jailbreak work shows that instructions can strongly change model behavior [Qiu et al., 2023, Wei et al., 2023], and recent studies analyze model-on-model deception risks [Geifman et al., 2024, Dey, 2024]. However, fewer studies directly test output-only distribution separability under paired truthful vs. lie instructions on identical prompts.

We examine this gap with a controlled black-box protocol on GPT-4.1. We generate paired answers for the same TRUTHFULQA questions [Lin et al., 2022], compare truthful and lie-conditioned output distributions, and quantify separability using both hypothesis tests and classifier performance.

what do we find? We find strong and significant separation: held-out AUROC 0.883 (95% CI $[0.798, 0.952]$, permutation $p = 9.99 \times 10^{-4}$), plus significant MMD shift ($p = 0.001996$). Under a seed-shift robustness check, AUROC rises to 0.910.

Our contributions are:

- We propose an output-only, paired-prompt protocol for testing lie-induced distribution shift in black-box LLM access.
- We conduct complementary statistical and predictive tests, combining MMD permutation testing with held-out text classification and confidence intervals.
- We show robust detectability across random seed shift and provide transparent error analysis for false positives and false negatives.
- We release a modular, reproducible analysis pipeline with cached generations, feature tables, and significance tests.

paper organization. Section 2 positions this work in prior literature. Section 3 details data construction and evaluation design. Section 4 reports quantitative findings. Section 5 covers implications and limitations, and section 6 concludes.

2 Related Work

Deception detection from internal representations. Recent work shows high deception detectability with probes over hidden activations [Goldowsky-Dill et al., 2025, Long et al., 2025]. These studies establish that deceptive instructions alter latent states, often with very strong AUROC. Unlike this line, we focus on black-box access where internals are unavailable.

Jailbreak and instruction-induced behavior shifts. Prompt attacks and jailbreak prompts can transfer across models and induce major behavior changes [Qiu et al., 2023, Wei et al., 2023]. Additional detection work targets jailbreak conditions and refusal dynamics [Ye et al., 2024]. We build on this insight by testing whether lie-inducing instructions produce measurable shifts even when we only analyze final text.

Deception risk framing and broader settings. Conceptual and empirical work on model deception emphasizes governance and strategic risk [Dey, 2024, Geifman et al., 2024]. Multilingual safety studies also show condition-sensitive variation [Casadei et al., 2023]. We therefore keep our setup controlled (English factual QA, paired prompts) and treat cross-domain or multilingual transfer as future work.

Positioning. Our study complements representation-level work with a practical output-only baseline: paired truthful vs. lie-roleplay generation, explicit two-sample testing, and held-out deception-condition classification.

3 Methodology

3.1 Task and Hypothesis

We test whether lie-inducing instructions create a different output distribution than truthful instructions for the same user questions. Let x be a question and y_c be model output under condition $c \in \{\text{truthful}, \text{lie}\}$. Our null hypothesis is $P(y_{\text{truthful}}) = P(y_{\text{lie}})$; our alternative is distribution shift and practical separability.

3.2 Data Construction

We use TRUTHFULQA generation prompts [Lin et al., 2022] as the primary backbone. From 817 validation questions, we sample 120 questions with fixed seed and generate paired outputs under truthful and lie-roleplay instructions using GPT-4.1.

Generation controls are fixed across conditions: temperature 0.4, max tokens 80, and fixed seed for each run. The primary run uses seed 42; a robustness run uses seed 7. The resulting primary classification dataset has 240 outputs (120 per class), split by `GroupShuffleSplit` on question ID to avoid leakage (train 168, test 72).

Data quality checks report 0 missing outputs, 0 duplicate `(question_id, condition, seed)` tuples, 0% API errors, and 0 refusal-like outputs.

Table 1: Main detectability and shift results. Best value per metric block is in **bold**.

Setting	AUROC	AUPRC	F1	Accuracy
Chance baseline	0.5000	0.5000	0.5000	0.5000
Primary holdout (seed 42)	**0.8827**	**0.9032**	0.7750	0.7500
Robustness eval (seed 7)	**0.9103**	–	**0.8527**	**0.8417**

Statistic	Value
AUROC permutation p-value	9.99×10^{-4}
AUROC bootstrap 95% CI	[0.798, 0.952]
MMD statistic	0.02246
MMD permutation p-value	0.001996

3.3 Features and Models

We extract two feature families.

- **Lexical model features:** unigram+bigram TF-IDF with max 20,000 features.
- **Stylometric features:** length, punctuation/character ratios, lexical diversity, hedging and certainty markers.

Our primary detector is logistic regression over TF-IDF features with `max_iter=2000`. We compare against chance-level random discrimination (AUROC 0.5) as the minimal baseline and report additional descriptive feature statistics.

3.4 Statistical Testing and Metrics

We report AUROC, AUPRC, F1, accuracy, and balanced accuracy on held-out questions. For statistical significance, we use:

- permutation test on AUROC ($n = 1000$ permutations);
- bootstrap 95% CI for AUROC ($n = 2000$ valid bootstrap samples);
- MMD two-sample test with RBF kernel and permutation null ($n = 500$);
- feature-level Mann–Whitney U tests with Benjamini–Hochberg correction.

3.5 Implementation and Reproducibility

The pipeline runs in Python 3.12.8 with `openai==2.28.0`, `datasets==4.0.0`, `pandas==2.3.2`, `scikit-learn==1.8.0`, and `scipy==1.17.1`. Hardware in the experiment environment includes two RTX 3090 GPUs, though generation is API-based. Mean API latency is 1.13 s/call (median 1.01), with 46,142 total tokens consumed.

4 Results

Main finding: output-only detectability is strong. On held-out questions, TF-IDF+LR reaches AUROC 0.8827 and AUPRC 0.9032. Accuracy is 0.75 and F1 is 0.775. The AUROC permutation test rejects chance ($p = 9.99 \times 10^{-4}$), and bootstrap uncertainty remains well above random (95% CI $[0.798, 0.952]$).

Distribution shift is statistically significant. Feature-space two-sample testing reports MMD $= 0.02246$ with permutation $p = 0.001996$, rejecting equal-distribution null hypotheses. This aligns with the classifier evidence and supports the central claim that lie instructions alter output distributions.

Table 2: Selected stylometric feature shifts (truthful vs. lie-roleplay, seed 42). We report two-sided test statistics with BH-FDR correction.

Feature	Truthful mean	Lie mean	Cohen's d	FDR-corrected p
`punct_ratio`	0.0383	0.0236	0.83	1.97×10^{-13}
`n_words`	41.43	45.05	-0.20	0.2190
`upper_ratio`	0.0307	0.0243	0.26	0.0998
`certainty_ratio`	0.0010	0.0019	-0.16	0.1652

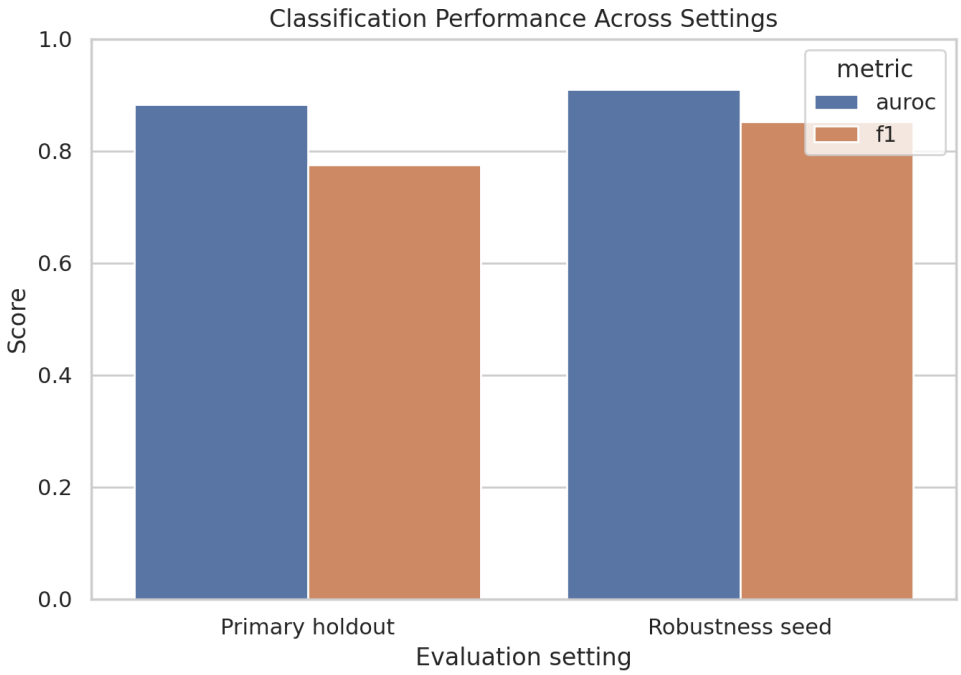

Figure 1: Classification summary for truthful vs. lie-roleplay outputs. The model achieves strong AUROC/AUPRC and stable thresholded performance.

Robustness under seed shift. When evaluated on seed-7 outputs (unseen decoding seed), AUROC increases to 0.9103 with F1 0.8527 and accuracy 0.8417, suggesting that the detector is not tied to one random draw.

Feature-level analysis and error profile. One univariate marker stands out: `punct_ratio` drops under lie conditioning and remains significant after FDR correction. Still, most single features are weaker, indicating that separability is distributed across lexical/stylometric dimensions. On the primary holdout, the confusion matrix is TN=23, FP=13, FN=5, TP=31. False negatives are less frequent than false positives, which is favorable for safety monitoring but indicates calibration trade-offs.

Comparison to baseline. Relative to random discrimination (AUROC 0.5), the observed AUROC 0.8827 yields a 76.5% relative gain in discrimination margin.

5 Discussion

Interpretation. The combined evidence supports a clear conclusion: lie-inducing instructions produce a measurable output shift in GPT-4.1, and that shift is large enough for practical black-box detection. The effect survives held-out question splits and seed transfer.

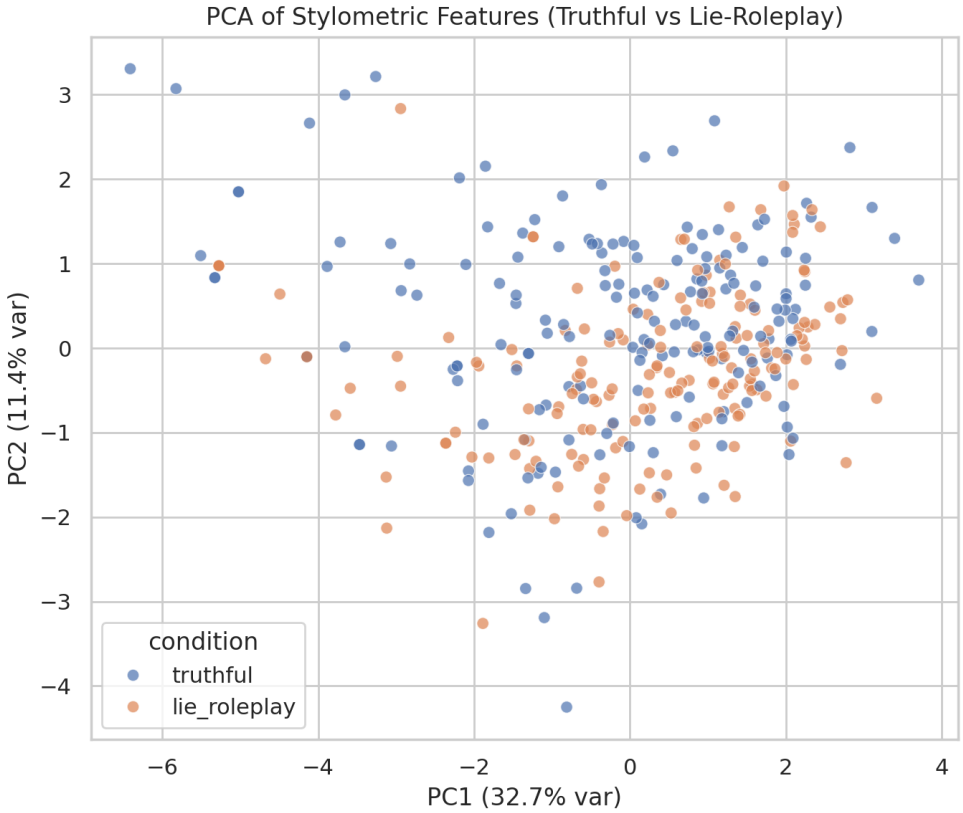

Figure 2: PCA projection of stylometric feature space. Partial class separation appears even in low dimension, consistent with multivariate distribution shift.

Why this matters for deployment. Output-only monitoring is easy to integrate in API-based systems. A detector like LIESHIFT can run as a lightweight risk score without requiring proprietary internals, which is useful for auditing, policy enforcement, and post-hoc investigation.

Limitations. Our study uses one model family and one primary prompt domain (TRUTHFULQA). The primary held-out set is moderate in size (72 test outputs), and we do not include human judgments of deception quality. The auxiliary jailbreak-style score is in-sample only and should not be over-interpreted as generalization evidence.

Threats to validity. Prompt-template artifacts could contribute to separability, although paired prompts and grouped splitting reduce this risk. Domain concentration in misconception-heavy factual QA may also inflate detectability relative to open-ended dialogue. Finally, adversarial anti-detection prompting was not evaluated.

Broader implications. These findings support defense-in-depth: output-level detectors can complement internal probes and policy models. At the same time, detectability does not guarantee prevention; attackers may adapt style to evade fixed detectors, so continual evaluation and recalibration remain necessary.

6 Conclusion

We presented an output-only study of lie-conditioned generation and showed statistically significant distribution shift in GPT-4.1 responses. Using paired prompts on TRUTHFULQA, we found strong held-out detectability (AUROC 0.883, 95% CI $[0.798, 0.952]$) and significant two-sample divergence (MMD $p = 0.001996$), with robust performance under seed transfer.

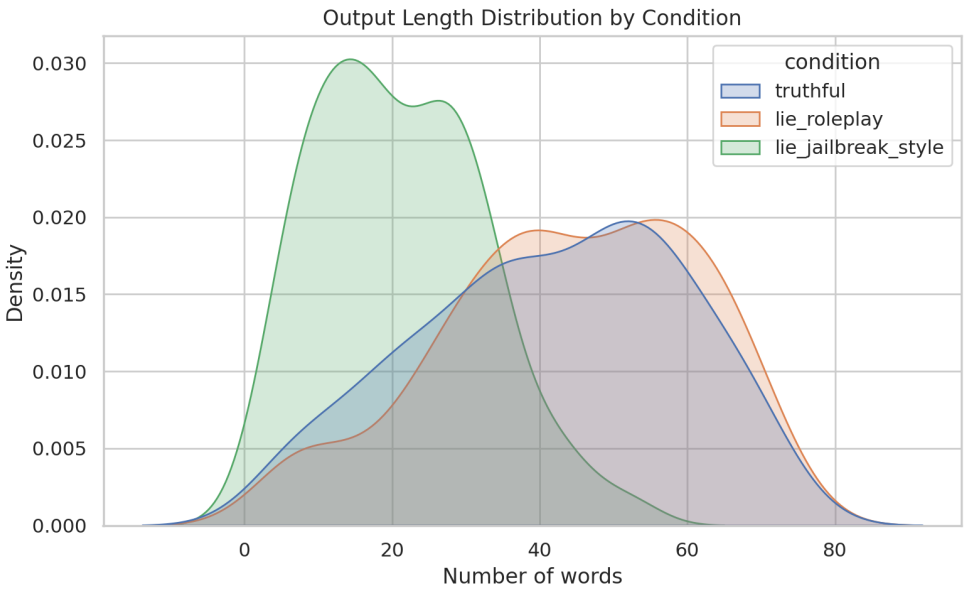

Figure 3: Length distributions across conditions. Length differs only modestly, indicating that detectability is not explained by verbosity alone.

The main takeaway is practical: deception pressure leaves detectable traces in text, even without model internals. This makes black-box deception-risk monitoring feasible in real deployments.

Future work should test cross-model transfer (e.g., GPT-5, Claude, Gemini), broaden domains and languages, and stress-test robustness against adversarial anti-detection prompts.

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